

Math 118 – Homework 1 Answers

HOMEWORK 1
due Wed, Sep. 9, 11:59 PM
(Upload your solutions to Gradescope)

1. (2 points) prove that a norm on a vector space V satisfies

$$|\|u\| - \|v\|| \leq \|u - v\| \text{ for all } u, v \in V.$$

Do not assume the norm comes from an inner product.

2. (4 points) (#1 p. 34.) Verify that the function

$$\langle Z, W \rangle = \sum_{j=1}^n Z_j \overline{W_j}$$

for $Z = (Z_1, \dots, Z_n)$, $W = (W_1, \dots, W_n) \in \mathbb{C}^n$ defines an inner product on $\mathbb{C}^n$ (i.e., satisfies Definition 0.1).

3. (2 points) (#3 p. 35.) Define $\langle V, W \rangle$ for $V = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and $W = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \in \mathbb{C}^2$ as

$$(\overline{w_1}, \overline{w_2}) \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

Show that $\langle V, V \rangle = 0$ for all vectors $V = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ with $v_1 + 2v_2 = 0$. Does $\langle \cdot, \cdot \rangle$ define an inner product?

4. (4 points) (#9 p. 35. Find an orthonormal basis for the orthogonal complement.)

Compute the orthogonal complement of the subspace in $\mathbb{R}^3$ spanned by the vector $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

5. (2 points) (#10 p. 35.) Let $f(t) = 1$ on $0 \leq t \leq 1$. Show that the orthogonal complement of f in $L^2[0, 1]$ is the set of all functions whose average value is zero.

6. (2 points) (#11 p. 35.) Show that if a differentiable function, $f(t)$, is orthogonal to $\cos(t)$ in $L^2[0, \pi]$, then $f'(t)$ is orthogonal to $\sin(t)$ in $L^2[0, \pi]$. *Hint: integrate by parts.*

7. (2 points) (#17 p. 36.) Suppose u_0 and u_1 are vectors in the inner product space V with $\langle u_0, v \rangle = \langle u_1, v \rangle$ for all $v \in V$. Show that $u_0 = u_1$. *Hint: Let $v = u_0 - u_1$.*

8. (2 points) (#23 p. 37.) Show that a set of orthonormal vectors is linearly independent.

Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggess and Narcowich.

Solutions

Problem 1

Solution notes

By the triangle inequality,

$$\|u\| = \|(u - v) + v\| \leq \|u - v\| + \|v\|$$

and, since $\|v - u\| = \|u - v\|$ by absolute homogeneity,

$$\|v\| = \|(v - u) + u\| \leq \|u - v\| + \|u\|.$$

Rearranging these two gives

$$-\|u - v\| \leq \|u\| - \|v\| \leq \|u - v\|,$$

which is what we wanted.

Problem 2

Solution notes

Since commutativity holds over addition and multiplication,

$$\langle u, v \rangle = \sum_{i=1}^n u_i \overline{v_i} = \overline{\sum_{i=1}^n v_i \overline{u_i}} = \overline{\langle v, u \rangle}.$$

Linearity on the first term and on the second term:

$$\langle au + bw, v \rangle = \sum_{i=1}^n (au_i + bw_i) \overline{v_i} = a \langle u, v \rangle + b \langle w, v \rangle, \quad (1)$$

$$\langle u, av + bw \rangle = \sum_{i=1}^n u_i (\overline{a v_i} + \overline{b w_i}) = \overline{a} \langle u, v \rangle + \overline{b} \langle u, w \rangle. \quad (2)$$

And for positive definiteness, $\langle u, u \rangle = \sum_{i=1}^n |u_i|^2 \geq 0$, which is 0 only when $u = 0$.

So this is an inner product on $\mathbb{C}^n$.

Problem 3

Solution notes

With $v_2 = -\frac{1}{2}v_1$,

$$\langle V, V \rangle = (\overline{v_1}, \overline{v_2}) \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = (\overline{v_1}, \overline{v_2}) \begin{pmatrix} v_1 + 2v_2 \\ 2v_1 + 4v_2 \end{pmatrix} \quad (3)$$

$$= (\overline{v_1}, \overline{v_2}) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0. \quad (4)$$

So $\langle V, V \rangle = 0$ whenever $v_1 + 2v_2 = 0$, for example $V = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \neq 0$. So positive definiteness fails, and this is not an inner product.

Problem 4

Solution notes

Compute the null space:

$$\mathbf{x} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0 \quad (5)$$

$$x - 2y + z = 0 \quad (6)$$

$$x = 2y - z \quad (7)$$

Thus,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore,

$$W^\perp = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Problem 5

Solution notes

Since $f(t) = 1$ on $[0, 1]$,

$$\langle f, g \rangle = \int_0^1 g(t) dt,$$

which is 0 exactly when g has average value 0 on $[0, 1]$ (the average is just $\frac{1}{1-0}$ times this integral).

So $f^\perp$ is the set of functions with average value zero.

Problem 6

Solution notes

Integrating by parts and using $\int_0^\pi f(t) \cos(t) dt = 0$,

$$\langle f', \sin(t) \rangle = [f(t) \sin(t)]_0^\pi - \int_0^\pi f(t) \cos(t) dt \quad (8)$$

$$= 0 - 0 = 0, \quad (9)$$

since $\sin(0) = \sin(\pi) = 0$.

So f' is orthogonal to $\sin(t)$ in $L^2[0, \pi]$.

Problem 7

Solution notes

Suppose $\langle u_0, v \rangle = \langle u_1, v \rangle$ for all $v \in V$. Take $v = u_0 - u_1$ and subtract, using linearity:

$$\langle u_0 - u_1, u_0 - u_1 \rangle = \langle u_0, u_0 - u_1 \rangle - \langle u_1, u_0 - u_1 \rangle = 0.$$

By positive definiteness this only happens when $u_0 - u_1 = 0$, so $\boxed{u_0 = u_1}$.

Problem 8

Solution notes

Let $\{u_1, \dots, u_n\}$ be orthonormal, so $\langle u_i, u_j \rangle = 0$ for $i \neq j$ and $\langle u_j, u_j \rangle = 1$. If $c_1 u_1 + \dots + c_n u_n = 0$,

take the inner product with u_j and use linearity:

$$0 = \sum_{i=1}^n c_i \langle u_i, u_j \rangle = c_j.$$

So every $c_j = 0$ and the vectors are linearly independent. An infinite set counts as linearly independent when every finite subset is, and any finite subset of the orthonormal set is orthonormal, so it is linearly independent.